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ORIGINAL ARTICLE

Accelerating equity: The causal impact of syntactic bootstrapping on inclusive education and SDG 4 for autistic children

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Abstract

To address critical challenges in inclusive education for autistic children under Sustainable Development Goal 4 (SDG 4), this study examined the causal impact of a syntactic bootstrapping intervention on language development and classroom participation. Using a quasi-experimental design, 18 autistic children aged 5–11 received a 12-week intervention and were compared with a control group receiving standard instruction. The results demonstrated a highly significant improvement in syntactic ability in the experimental group ($M = 18.7$, SD = 3.2) compared with the control group ($M = 11.4$, SD = 4.1; $p < 0.001$). These linguistic gains functioned as a catalyst for educational inclusion, as evidenced by a strong positive correlation between syntactic development and active classroom participation ($r = 0.78$, $p < 0.001$), with regression analysis confirming syntactic ability as a primary predictor of inclusive participation ($\beta = 0.65$, $p < 0.001$). Beyond its empirical contribution, the study highlights the educational relevance of syntactic development for supporting participation in inclusive classrooms. The findings suggest that targeted syntactic instruction, when embedded in classroom practice, may enhance engagement and learning autonomy among autistic students. While context-specific, the results contribute classroom-based evidence relevant to inclusive education efforts aligned with SDG 4.

KEYWORDS
autism, inclusive education, language intervention, sustainable development goals (SDG 4), syntactic bootstrapping

Key Points

- A 12-week syntactic bootstrapping intervention significantly improved the syntactic abilities of autistic children compared with standard classroom instruction in Indonesian inclusive schools.
- Improved syntactic ability was positively associated with greater classroom participation, including communication, engagement, social interaction and reduced reliance on adult prompting.
- Embedding language-focused intervention within everyday classroom instruction may support autistic students' participation while strengthening teachers' roles as co-implementers of inclusive educational practices.
- The findings provide context-specific evidence from an Indonesian inclusive education setting and highlight the potential contribution of language-based interventions to educational participation aligned with SDG 4.

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INTRODUCTION

Sustainable Development Goal 4 (SDG 4) positions inclusive education as a global commitment to ensuring equitable access to quality education for all learners, including children with autism spectrum disorder (United Nations, 2015). Children with autism spectrum disorder (ASD) have persistent challenges with social communication and interaction, alongside restricted interests and repetitive behaviours, impacting their ability to function (Yan et al. 2024; Wang & Wang, 2024). Within this framework, inclusive education is commonly defined as the provision of meaningful learning opportunities for students with diverse needs within regular educational settings, supported by appropriate pedagogical, curricular and social adaptations (Guillén-Martínez et al., 2025; UNESCO, 2023). For autistic children, inclusion is not merely physical placement in mainstream classrooms but active participation in learning activities, peer interaction and communicative engagement (Odom & Et, 2020; Yang et al., 2025).

International research conducted predominantly in Western and high-resource educational systems has identified language ability, particularly syntactic competence, as a key factor influencing autistic children's participation in inclusive classrooms (Roberts & Al, 2019; Smith & Al, 2020). These studies demonstrate that syntactic development supports comprehension of instructional language, initiation of social interaction and learning autonomy. However, the direct applicability of these findings to low- and middle-income countries (LMICs) remains uncertain, as educational infrastructure, teacher preparation and support services differ substantially across contexts (Diny Sujannah et al., 2025; Divan & Al, 2021).

In Indonesia, inclusive education is formally mandated through Minister of National Education Regulation No. 70 of 2009. Despite this regulatory commitment, implementation remains uneven, particularly for autistic learners (Setiawan & Tohari, 2025). National data indicate that only a small proportion of inclusive schools provide structured language interventions tailored to autism, and most teachers report limited access to specialized training or multidisciplinary support (Sunardi & Yusuf, 2019; Kemendikbudristek, 2023). As a result, autistic children frequently experience restricted classroom participation, limited peer interaction and dependence on adult prompting, even when enrolled in inclusive schools (Mariyam, 2025; Puspitasari & Al, 2022).

It is important to note that the Indonesian educational context differs markedly from the Western settings in which most syntactic intervention studies have been conducted. Indonesian classrooms typically operate within larger class sizes, limited specialist support and an agglutinative language structure that poses distinct linguistic processing demands (Puspitasari & Al, 2022). Therefore, evidence derived from high-resource contexts cannot be assumed to transfer directly without contextual validation. This contextual gap highlights the need for locally grounded empirical research that examines whether established language interventions can function effectively within Indonesian inclusive education settings.

From a theoretical perspective, participation in inclusive education is understood within special-education theory as a multidimensional construct encompassing behavioural engagement, social interaction, communicative contribution and learning autonomy (Odom & Et, 2020; Wang et al., 2024). Language ability, particularly syntactic competence, is theorized to function as a foundational mechanism enabling these dimensions of participation. Syntactic bootstrapping, which leverages children's lexical knowledge to infer grammatical structures, has been widely recognized as a promising approach for supporting language development in autistic children (Lailiyah & Gigik, 2024; Naigles, 2020). Nevertheless, existing studies have largely focused on linguistic outcomes alone, with limited examination of how syntactic gains translate into observable participation in inclusive classroom settings, especially in LMIC contexts.

Taken together, the literature reveals three interrelated gaps. First, there is a lack of causal evidence linking syntactic intervention to inclusive classroom participation, as many prior studies rely on correlational or single-case designs. Second, there is a contextual gap, as intervention research is heavily concentrated in Western, high-resource educational systems, leaving LMIC settings underrepresented. Third, there is a conceptual gap in connecting language intervention outcomes to inclusive education constructs aligned with SDG 4, such as equity, access and quality participation.

In response to these gaps, the present study aims to examine the causal impact of a syntactic bootstrapping intervention on both syntactic development and participation in inclusive education among autistic children in Indonesia. Specifically, this study seeks to investigate whether improvements in syntactic ability contribute to enhanced engagement, communication, social interaction and learning autonomy in inclusive classroom contexts. Through its quasi-experimental design, this study tests the hypothesis that the syntactic bootstrapping intervention not only enhances linguistic competence but also acts as a catalyst for equitable access to quality education, aligning with the 'multiplier effect' strategy within SDG 4 (UNICEF, 2022). The findings are expected to form the basis for transformative policy recommendations for sustainable, inclusive education in Indonesia.

Accordingly, the study addresses the following research questions:

1. Does a syntactic bootstrapping intervention significantly improve the syntactic abilities of autistic children in Indonesian inclusive education settings?

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2. To what extent do gains in syntactic ability predict autistic children's participation in inclusive classrooms, as measured across engagement, communication, social interaction and learning autonomy?

METHOD

Research design and rationale

This study employed a quasi-experimental, non-equivalent control group design to examine the effect of a syntactic bootstrapping intervention on syntactic development and inclusive classroom participation among autistic children (Creswell, 2023). Random assignment was not feasible due to ethical and institutional constraints inherent in inclusive school settings, where classroom composition and parental consent must be respected. Accordingly, the design follows established practices in applied educational and special-education research, where causal inference is pursued under real-world conditions rather than laboratory control. The quasi-experimental approach was selected to balance methodological rigour with ecological validity, allowing the intervention to be implemented within authentic inclusive classroom contexts. While this design limits statistical generalizability, it is appropriate for examining intervention effects in educational environments where randomization may disrupt students' learning continuity.

Participants, setting and recruitment

Participants were recruited from three public inclusive elementary schools in Kediri City, Indonesia, that formally implement inclusive education programs under national policy. School principals and inclusive education coordinators facilitated recruitment following institutional approval. Written informed consent was obtained from parents or legal guardians prior to participation. The final sample consisted of 35 autistic children aged 5–11 years, with 18 assigned to the experimental group and 17 to a comparison group receiving standard classroom instruction. Participants were selected using purposive sampling based on the following criteria: (a) a clinical diagnosis of Autism Spectrum Disorder (Level 1–2) by a licensed child psychiatrist according to DSM-5-TR criteria; (b) Childhood Autism Rating Scale (CARS) scores between 30 and 36.5, indicating mild-to-moderate autism; and (c) documented syntactic delays as identified by pre-intervention screening. The comparison group was drawn from similar inclusive classrooms within the same schools to ensure contextual equivalence in curriculum, teacher qualifications and learning environment. This non-random allocation reflects ethical constraints typical of inclusive educational research.

Educational context and teacher role

The intervention was implemented within regular inclusive classroom settings rather than clinical or therapeutic environments. Classroom teachers served as co-implementers of the intervention following structured training sessions conducted by the research team. Teachers integrated syntactic bootstrapping activities into daily instructional routines, including literacy lessons, collaborative tasks and guided play. This pedagogical integration was designed to ensure that language development occurred within meaningful educational interactions rather than isolated drills. Teacher involvement was central to the intervention, reflecting the study's alignment with special-education practice and inclusive pedagogy.

Intervention and implementation fidelity

The syntactic bootstrapping intervention was delivered over 12 weeks in 24 sessions (two sessions per week, 45 min per session). Instructional materials were adapted to the Indonesian linguistic and cultural context using age-appropriate folk narratives and *wayang*-based visual scaffolding. Implementation fidelity was monitored through structured observation checklists and session logs completed by trained observers. Inter-rater reliability for fidelity coding reached $\kappa = 0.88$, indicating high consistency. These procedures ensured that instructional delivery remained aligned with the intervention protocol across classrooms. Figure 1 presents the data collection which was conducted in the following three stages:

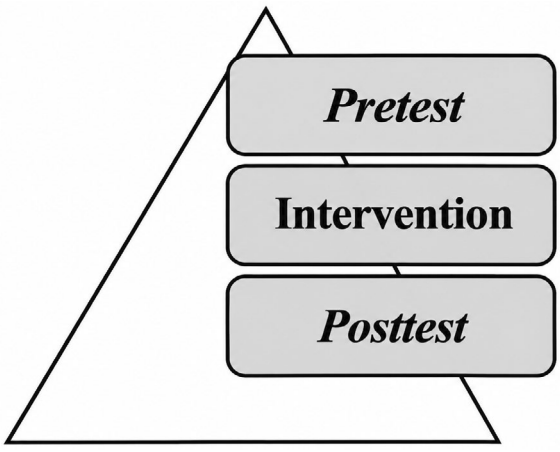


FIGURE 1 Data collection stages.

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Measures and conceptual justification

Syntactic ability

Syntactic development was measured using the Indonesian adaptation of the Test of Early Grammatical Impairment (TEGI-BI). The adaptation process involved linguistic equivalence review by bilingual language experts, pilot testing with Indonesian-speaking autistic children and modification of items to reflect agglutinative morphosyntactic features of Bahasa Indonesia. The instrument demonstrated high internal consistency ($\alpha = 0.91$). TEGI-BI was selected because syntactic competence is theoretically linked to instructional comprehension and communicative participation in inclusive classrooms.

Inclusive classroom participation

Inclusive participation was measured using the Inclusive Classroom Observation Scale (ICOS), which operationalizes participation across four educational dimensions: engagement, communication, social interaction and learning autonomy. These dimensions align with established special-education frameworks defining meaningful inclusion beyond physical placement. ICOS scores were used as educational proxies for inclusive participation rather than macro-level SDG indicators.

Teacher reports and video data integration

Teacher reports were collected using structured participation checklists aligned with the four domains of the Inclusive Classroom Observation Scale (ICOS): engagement, communication, social interaction and learning autonomy. Classroom video recordings were analysed using a structured observational coding protocol based on these same predefined indicators to ensure conceptual consistency across data sources. Coding focused on observable behaviours, including the frequency of peer-initiated interactions, responsiveness to teacher instructions, participation in collaborative activities and reliance on adult prompts.

Two members of the research team who had received prior training in the coding protocol independently coded a randomly selected subset (30%) of the video recordings before completing the remaining observations. Coding agreement was evaluated using Cohen's kappa, resulting in substantial inter-rater agreement ($\kappa = 0.86$). Any coding discrepancies were resolved through discussion until consensus was reached, after which one trained coder completed the remaining observations using the agreed coding framework.

Teacher reports and video observations were not analysed as independent qualitative datasets. Instead, they were used as confirmatory sources to triangulate the quantitative ICOS findings and to provide contextual evidence regarding changes in classroom participation following the intervention. Patterns that converged across teacher reports, observational coding and ICOS scores were used to strengthen the interpretation of the intervention effects.

Ethics statement

The study received ethical approval from the relevant institutional review board and authorization from participating schools. All procedures complied with ethical standards for educational research involving children. Parents or guardians provided written informed consent, and participant anonymity and data confidentiality were strictly maintained.

Sample size and analytical consideration

Given the constraints of recruitment within inclusive school settings, the sample size was necessarily limited. To evaluate whether the sample was sufficient to detect the observed intervention effect, a post-hoc power analysis was conducted using G*Power version 3.1.9.7 (Faul et al., 2009). The analysis was based on the primary ANCOVA model, with a total sample size of 35 participants (18 experimental, 17 control), a significance level of $\alpha = 0.05$, and the observed effect size (partial $\eta^2 = 0.57$, equivalent to Cohen's f = 1.15). The analysis yielded an achieved statistical power of $1 - \beta > 0.99$, indicating that the study was adequately powered to detect the intervention effect. In accordance with recommendations for quasi-experimental educational research, analytical emphasis was placed on effect size estimation and the consistency of findings rather than statistical significance alone. Statistical procedures were selected to address the research questions while acknowledging the inherent limitations of the study design.

Data analysis

Data analysis was conducted to address the study's research questions concerning (a) the effectiveness of the syntactic bootstrapping intervention on syntactic development and (b) the relationship between syntactic ability and participation in inclusive classrooms. Analytical decisions were guided by the quasi-experimental design and the educational focus of the study rather than by purely statistical considerations. All quantitative analyses were performed using IBM SPSS Statistics (Version 29). Preliminary analyses included descriptive statistics (means, standard deviations and distribution checks) to assess data

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quality and baseline equivalence between groups. Independent-samples *t*-tests were used at pretest to examine initial group comparability, acknowledging the non-randomized nature of group assignment.

To examine intervention effectiveness while controlling for baseline differences, an Analysis of Covariance (ANCOVA) was conducted, with posttest syntactic scores as the dependent variable and pretest scores entered as a covariate. This approach is widely recommended in educational intervention research to strengthen causal inference in non-equivalent group designs. Effect sizes (partial η^2 and Cohen's d) were calculated to support educational interpretation beyond statistical significance. Moreover, to examine changes over time within the experimental group, repeated-measures analyses were conducted to identify developmental patterns across the intervention period. Clinical significance was assessed using the Reliable Change Index (RCI) to determine whether observed gains reflected meaningful individual improvement rather than measurement error.

To address the second research question, Pearson correlation analysis was used to examine the association between syntactic development and inclusive classroom participation. Multiple linear regression analysis was then employed to estimate the predictive contribution of syntactic ability to participation outcomes, while controlling for age. These analyses were selected to examine theoretically grounded relationships between language development and educational participation rather than to generate purely statistical models.

Teacher reports and video-based participation coding were summarized descriptively and compared with ICOS outcomes to examine convergence across data sources. Rather than being subjected to independent qualitative thematic analysis, these data served as triangulation evidence that contextualized the quantitative findings. Specifically, observational patterns derived from the coding process were used to verify whether improvements identified through ICOS scores were reflected in naturally occurring classroom behaviours, including increased peer interaction, greater engagement with instructional activities and reduced dependence on adult prompting. Analytical emphasis was therefore placed on convergence between observational evidence and statistical outcomes to strengthen the credibility and ecological validity of the study's interpretations. The data analysis process is illustrated in Figure 2.

RESULTS

The results of the study highlight the findings from the data analysis, organized into three main sections to build a comprehensive picture of the intervention's impact: (1) the Effectiveness of the Syntactic Bootstrapping Intervention on the core syntactic abilities of the autistic children; second, (2) the consequential Impact on Participation in Inclusive Education (ICOS) following the intervention, linking linguistic gains to tangible improvements in the classroom and (3) outcomes synthesized to demonstrate the study's broader Contribution to key Sustainable Development Goal 4 (SDG 4) Indicators.

RQ1: Does the syntactic bootstrapping intervention significantly improve the syntactic abilities of autistic children in inclusive education settings?

To address the first research question, descriptive statistics were examined to compare syntactic performance between the experimental and control groups before and after the intervention. Table 1 summarizes the pretest and posttest TEGI-BI scores of the 18 participants in the experimental group, including means, standard deviations, gain scores and gain ranges.

Table 1 shows higher TEGI-BI scores at posttest than at pretest following the intervention. The mean pretest score was 12.3 (SD = 2.1), whereas the mean posttest score rose to 18.7 (SD = 3.2). The mean gain score was +6.4 (SD = 1.8). Gain scores ranged from +3 to +9, with all

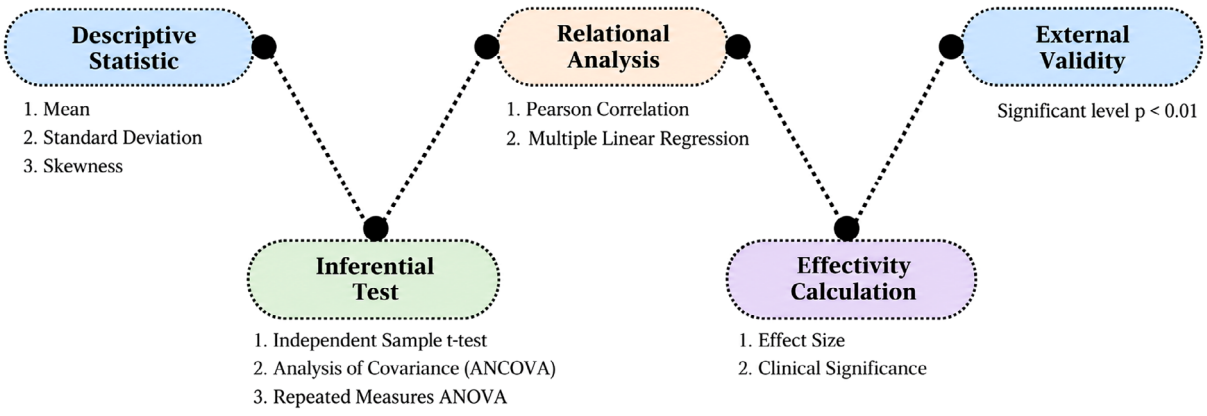


FIGURE 2 Data analysis.

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TABLE 1 Statistical analysis and clinical significance of the intervention.

Parameter	Pretest ($M \pm SD$)	Posttest ($M \pm SD$)	Gain score ($M \pm SD$)	Gain range
TEGI-BI Score	12.3±2.1	18.7±3.2	+6.4±1.8	[+3 to +9]

TABLE 2 Comparison with the control group.

Indicator	Experimental ($n=18$)	Control ($n=17$)	Δ	p-value
Mean syntactic gain	+6.4	+1.2	+5.2	<0.001
MCID achievement (≥5 points)	86%	12%	+74%	<0.001

participants showing positive gains. Overall, the experimental group demonstrated higher posttest TEGI-BI scores than pretest scores.

ANCOVA, using pretest scores as the covariate, showed statistically significant differences between the experimental and control groups, as presented in Table 2. Table 2 presents a comparison of syntactic gains between the experimental and control groups following the intervention. The experimental group demonstrated a substantially higher mean syntactic gain (+6.4 points) than the control group (+1.2 points), resulting in a mean difference of +5.2 points. This between-group difference was statistically significant ($p<0.001$). In the experimental group, 86% of participants achieved the minimum clinically important difference (MCID) threshold of at least five points, compared with 12% in the control group, representing a 74% difference between groups ($p<0.001$).

Clinical significance was further examined using the Reliable Change Index (RCI) and the Minimal Clinically Important Difference (MCID). A total of 94% of participants (17 out of 18 children) achieved a Reliable Change Index (RCI) greater than 1.96. This value exceeded the conventional RCI threshold of 1.96. Furthermore, 86% of participants (15 out of 18 children) showed a minimum improvement of 5 points, meeting the standard for a Minimal Clinically Important Difference (MCID).

RQ2: To what extent do gains in syntactic ability predict autistic children's participation in inclusive classrooms?

The second research question examined inclusive classroom participation following the intervention. Based on the Inclusive Classroom Observation Scale (ICOS), Table 3 presents the posttest scores of the 18 participants in the experimental group across four domains: engagement, social interaction, communication and learning autonomy.

TABLE 3 Participant achievement.

ICOS domain	Mean score ($M \pm SD$)	Score range
Engagement	8.9±1.2	7–10
Social interaction	9.1±1.4	6–10
Communication	9.3±1.1	8–10
Learning autonomy	7.9±1.6	5–10
Total ICOS score	35.2±4.5	28–40

TABLE 4 Comparative analysis with the control group.

Parameter	Experimental ($n=18$)	Control ($n=17$)	Δ	p-value
Total ICOS score	35.2±4.5	28.7±5.1	+6.5	<0.001
Interaction frequency/ Hour	12.3±2.1	7.1±3.0	+5.2	<0.001
Prompt dependency	2.4±1.0	5.8±1.7	−3.4	<0.001

Based on the ICOS results, participants obtained mean scores across all four participation domains. The Engagement domain had a mean score of 8.9 (SD=1.2), with scores ranging from 7 to 10. The Social Interaction domain had a mean score of 9.1 (SD=1.4), with scores ranging from 6 to 10. The Communication domain had the highest mean score ($M=9.3$, SD=1.1), with scores ranging from 8 to 10. Meanwhile, the Learning Autonomy domain had a mean score of 7.9 (SD=1.6), with scores ranging from 5 to 10. The wider score range in the Learning Autonomy domain indicates greater variability among participants than in the other ICOS domains. The mean total ICOS score was 35.2 (SD=4.5), with observed scores ranging from 28 to 40. Among the four domains, Communication and Social Interaction had the highest mean scores, whereas Learning Autonomy had the lowest mean score.

To compare participation outcomes between groups, descriptive and inferential analyses were conducted. Table 4 reports differences in total ICOS scores, interaction frequency and prompt dependency between the experimental and control groups.

Table 4 compares inclusive classroom participation outcomes between the experimental and control groups. The experimental group obtained a higher mean total ICOS score (35.2±4.5) than the control group (28.7±5.1), yielding a between-group difference of 6.5 points, which was statistically significant ($p<0.001$). In addition, participants in the experimental group demonstrated a higher frequency of classroom interactions per hour (12.3±2.1) compared with the control group (7.1±3.0), corresponding to a mean difference of 5.2 interactions ($p<0.001$). Conversely, prompt dependency was lower in the experimental group (2.4±1.0) than in the control group (5.8±1.7), with a difference of −3.4 ($p<0.001$).

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Overall, the experimental group obtained higher ICOS scores and interaction frequencies and lower prompt dependency than the control group.

To examine the relationship between syntactic ability and participation, Pearson correlation and regression analyses were performed. As depicted in Table 5, the Pearson correlation analysis showed a very strong and significant positive relationship between these two variables ($r=0.78$, $p<0.001$), indicating that nearly 61% of the variance in social participation can be explained by syntactic progress. This consistent linear relationship suggests that every 1-point increase on the Test of Early Grammatical Impairment-Bahasa Indonesia (TEGI-BI) predictively correlates with an approximate 1.2-point increase on the Inclusive Classroom Observation Scale (ICOS). As a concrete illustration, participant E03, who experienced a dramatic 9-point increase on the TEGI-BI, showed a comparable 11-point increase in social participation on the ICOS, reinforcing the validity of this predictive relationship in an individual context.

Table 5 presents the results of the linear regression analysis. With a regression coefficient of $\beta=0.65$ ($p<0.001$), the model yields the predictive equation: ICOS Score = 12.37 + 0.65 (Syntax Score) + 0.12 (Age). Syntactic ability was positively associated with ICOS scores ($\beta=0.65$, $p<0.001$), whereas the regression coefficient for age was 0.12. The analysis of variance shows that syntactic ability alone accounts for 42.3% of the total variance in social participation ($R^2=0.423$). The model explained 42.3% of the variance in ICOS scores ($R^2=0.423$).

In relation to educational participation outcomes aligned with SDG 4 targets, teacher reports and classroom observations were used to support the interpretation of ICOS scores. Teachers completed structured participation checklists corresponding to ICOS domains, including engagement, communication, social interaction and learning autonomy. These reports were summarized descriptively and examined for consistency with observational scores rather than analyzed independently. Classroom video recordings were coded using predefined behavioural indicators (e.g., frequency of peer-initiated interaction, responsiveness to instructional language and reliance on adult prompts). Observational coding showed higher levels of engagement and interaction and lower prompt dependency in the experimental group than in the control group. These data are reported as classroom-level participation evidence relevant to inclusive education outcomes, not as direct measures of SDG attainment. The convergence between teacher reports, video observations and ICOS scores increased confidence that the observed statistical improvements reflected authentic changes in classroom participation rather than measurement artefacts.

DISCUSSION

Interpretation of findings

This study examined the relationship between a syntactic bootstrapping intervention, syntactic development and classroom participation among autistic children in inclusive education settings. Consistent with prior research on syntactic bootstrapping and language acquisition in autism (Göksun et al., 2020; Goodwin & Et, 2015; Naigles, 2020), the findings indicate that participants who received the intervention demonstrated greater syntactic gains than peers receiving standard instruction. These results align with earlier evidence that syntactic scaffolding can support grammatical development in autistic learners (Silleresi & Tuller, 2025; Wu & Cheng, 2024), while extending this work into an authentic classroom-based educational context.

Beyond syntactic outcomes, the study contributes to inclusive education research by demonstrating an association between syntactic development and classroom participation. This finding supports theoretical perspectives that conceptualize language as a foundational resource for learning engagement and social interaction (Prasetyo, 2024; Roberts & Al, 2019). Importantly, the present results should be interpreted as context-specific

TABLE 5 Correlation and regression of the syntax-participation relationship.

Analysis	Method/model	Main outcome	Effect size/ coefficient	Significance
Syntax-participation relationship	Pearson correlation	$r=0.78$; every +1 TEGI-BI ≈ +1.2 ICOS	$r=0.78$	
Observational example	—	Participant E03: ΔTEGI-BI = +9 → ΔICOS = +11	—	—
Prediction of syntactic contribution	Linear regression	Syntax $\beta=0.65$	$\beta=0.65$	$p<0.001$
Regression equation	ICOS Score = 12.37 + 0.65 (Syntax) + 0.12 (Age)	Intercept 12.37; Age Coef1 0.12	Intercept = 12.37; Age = 0.12	—
Variance explained	Coefficient of determination (R^2)	Syntax explains 42.3% of the variance participation	$R^2=0.423$	—

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and exploratory, reflecting patterns observed within a small, non-randomized sample rather than definitive causal relationships applicable across settings.

Mechanisms linking language intervention and classroom participation

The observed association between syntactic improvement and participation outcomes can be understood through several educational mechanisms. Improved syntactic competence may have enhanced children's ability to comprehend instructional discourse, particularly multi-clause or sequential instructions commonly used in classroom teaching. This is consistent with evidence that syntactic processing plays a key role in academic readiness and task engagement for autistic learners (Lailiyah & Gigik, 2024; Goodwin & Et, 2015). At the expressive level, increased grammatical competence may support clearer initiation of interaction and more sustained participation in peer exchanges, as suggested by prior classroom-based autism research (Odom & Et, 2020; Wolfer et al., 2024).

However, participation in inclusive classrooms is shaped by more than individual language ability alone. Classroom structure, teacher mediation, peer responsiveness and instructional design all interact with student competencies to influence participation opportunities (Foykas et al., 2025; Mohamed, 2025). Accordingly, syntactic development should be viewed as an enabling factor that supports participation when aligned with inclusive pedagogical practices (Timofeeva & Chugunova, 2025), rather than as a singular determinant of inclusion.

Contribution to SDG 4 indicators

Although SDG 4 outcomes cannot be measured directly within a classroom-based intervention, the observed improvements in participation provide educational evidence that is conceptually aligned with several SDG 4 targets, particularly equitable participation, inclusive learning environments and learning engagement.

Equitable access (target 4.5)

The improvements in access to adaptive materials and classroom participation suggest that syntactic intervention may reduce barriers to educational participation for autistic learners. According to Table 6, the frequency of access to adaptive materials reached an impressive level of 92%, a dramatic increase of 53% compared to the control group. This achievement suggests that the children were not only able to identify and utilize the communicative resources available in their environment but also developed the ability to proactively seek and use communication tools appropriate for their expressive needs. This high frequency may reflect a growing metacognitive awareness of alternative communication strategies and the ability to adapt to various available communication modalities.

TABLE 6 Equal access.

Indicator	Experimental group	Δ versus control
Frequency of access to adaptive materials	92%	+53%
Participation in class activities	89%	+47%
Peer interaction	12.3 interaction/h	+73%

Participation in class activities showed an equally significant increase, reaching 89%, which was 47% higher than the control group. This indicator suggests meaningful improvement in the children's motivation and self-confidence to engage in the dynamics of collective learning. The increase in class participation may indicate broader improvements not only in receptive and expressive language skills but also in the development of broader social skills, including the ability to understand group interaction norms, regulate conversational turns and respond appropriately to various communicative situations that arise in a formal learning context.

Increased peer interaction may represent the dramatic increase in peer interaction, which averaged 12.3 interactions per hour, a 73% increase compared to the control group. This surge in the quantity of interactions shows that the children not only progressed in their technical communication abilities but also became more willing to engage socially. This high frequency of interaction may reflect improvements in socio-pragmatic competence, including the ability to initiate conversations, maintain topics and navigate the social nuances of interacting with same-aged peers. This improvement is highly meaningful, as peer interaction provides a natural context for developing spontaneous communication skills and generalizing learned language abilities into authentic social situations.

Inclusive facilities (target 4.a)

An important educational outcome of this intervention was the dramatic transformation in the need for special aid support, which is directly related to Sustainable Development Goal Target 4, concerning the provision of inclusive and child-friendly educational facilities. The data show a remarkable 60% reduction in the need for a special aide. Before the intervention, 72% of the children required intensive special assistance to participate in regular learning activities; post-intervention, this figure dropped drastically to only 29% still requiring aide support. This decrease not only demonstrates an increase

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in the children's adaptive skills and individual independence but also indicates that their progress in syntactic and communication abilities has transformed their capacity to understand and respond to learning instructions independently.

Qualitative reports from teachers via the Teacher Report of Students' Performance (TRSP) provide deep insight into the mechanism behind this transformation. Teacher observations revealed that 'the child was able to use independent learning tools after understanding complex verbal instructions,' a statement that captures the essence of the fundamental change in the children's linguistic processing. The ability to understand complex verbal instructions shows marked improvement in receptive language ability, particularly in processing multi-layered syntactic structures and parsing linguistic information that contains multiple commands or conditional statements. This improvement allows the children to not only follow simple instructions but also to understand nuances in meaning and identify task priorities.

The practical consequence of this progress is the development of learning independence, wherein children can operate learning tools, sequence activities and perform self-monitoring of their progress without requiring constant intervention from an adult caregiver. This transformation may have practical implications for classroom support, not only reducing the burden on the educational system in terms of human resource allocation but also creating a more inclusive classroom environment where children on the autism spectrum can participate meaningfully in the learning community alongside their neurotypical peers.

Quality learning (target 4.1)

The intervention's impact on learning readiness showed a substantial improvement in preparing children to face more complex academic demands, aligning with Sustainable Development Goal Target 4.1, which emphasizes the importance of ensuring all children complete a quality primary education with relevant and effective learning outcomes. The data indicate a dramatic 22.6% increase in learning readiness within the experimental group. The proportion of children capable of handling complex tasks surged from only 38% in the pre-intervention phase to 86% being able to complete collaborative tasks post-intervention. This transformation demonstrates that syntactic progress not only impacts communication ability but also supports children's capacity.

The most striking illustration of this transformation can be seen in the case of participant E03, who underwent a substantial communicative improvement. Initially, this participant was only able to provide minimal 1-to-2-word responses, indicating significant limitations in expressive ability and likely difficulties with internal linguistic organization. Post-intervention, however, the same participant spontaneously, without stimulation or command, produced a complex sentence demonstrating greater use of multiple linguistic competencies: '*Kita warna mobil, nanti kumpulkan sama!*' (meaning 'Let's color the car, then we'll gather them together!'). This sentence not only shows an advance in syntactic structure through the use of coordinate clauses but also indicates evidence of improved pragmatic communication. This includes the ability to use inclusive pronouns ('*kita*' meaning 'we'), provide a structured temporal sequence ('*nanti*' meaning 'later') and facilitate a collaborative activity ('*kumpulkan sama*' meaning 'gather together'). Furthermore, the ability to lead a group discussion suggests the child has developed meta-cognitive awareness of group dynamics, leadership skills and the capacity to organize a collective learning activity. This transformation from a passive respondent to an active facilitator in collaborative learning shows that the syntactic intervention illustrates how syntactic improvement may facilitate more active participation in collaborative learning, allowing the child to become not just a recipient of information but also an active contributor to the co-construction of knowledge.

Implications for educational practice

From a pedagogical perspective, the findings suggest that language-focused interventions may be most effective when embedded within routine classroom instruction rather than delivered as isolated clinical activities. This aligns with inclusive education frameworks that emphasize participation through everyday learning interactions (Odom & Et, 2020). Teachers' involvement as co-implementers of the intervention highlights the importance of instructional practices that integrate language scaffolding into literacy activities, collaborative tasks and guided play.

These implications are offered cautiously and without prescriptive intent. As noted in prior work on inclusive intervention research in low- and middle-income contexts (Divan & Al, 2021; Prasetyo, 2024), implementation feasibility depends heavily on teacher workload, classroom conditions and available support systems. Therefore, the present findings should be understood as informing reflective practice rather than prescribing uniform instructional models.

Cultural, pedagogical and ethical considerations

The study also raises important cultural and ethical considerations relevant to inclusive education research. Adapting syntactic bootstrapping materials to local

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linguistic and cultural contexts reflects recommendations that interventions be culturally responsive rather than directly transplanted from high-resource Western settings (Divan & Al, 2021; Goodwin & Et, 2015). Such adaptation is particularly relevant in agglutinative language contexts, where grammatical structures and instructional discourse differ from those examined in much of the existing literature.

Ethically, the study reflects common constraints in school-based research involving autistic children. The absence of random assignment and reliance on purposive sampling reflect ethical priorities related to parental consent, classroom stability and learner well-being (Creswell, 2023). These considerations reinforce the value of quasi-experimental designs in inclusive education research, where methodological rigour must be balanced against educational and ethical responsibilities.

Limitations and future research

Several limitations should be considered when interpreting the findings of this study. First, the relatively small sample size and the use of a quasi-experimental, nonrandomized design constrain the generalizability of the results. Although effect sizes were reported to support interpretation, the findings should be understood as context-specific and exploratory rather than indicative of population-level effects.

Second, the diversity of participants was limited, as the study focused exclusively on autistic children with mild to moderate symptom severity (CARS scores 30–36.5). This focus enabled clearer examination of intervention effects within a relatively homogeneous group but limits conclusions regarding applicability across the broader autism spectrum. Future research should extend this line of inquiry to include children with more severe autism profiles, examining whether syntactic bootstrapping remains effective under higher support needs and identifying the instructional adaptations required to accommodate differing communicative and cognitive profiles.

Third, while teacher reports and classroom observations were incorporated as confirmatory data sources, the study did not employ systematic qualitative analysis of classroom interaction. Future studies would benefit from mixed-methods designs that integrate qualitative approaches, such as classroom discourse analysis or teacher interviews, to capture how language intervention interacts with instructional practices, peer dynamics and classroom culture.

Finally, the present study examined short-term outcomes following a 12-week intervention. Longitudinal research is needed to assess the durability of syntactic and participation gains over time and across transitions between grade levels or instructional settings. Collectively, such future research would strengthen understanding of how language-based interventions can support inclusive education across diverse learner profiles and educational contexts.

CONCLUSION

This study examined the educational effects of a syntactic bootstrapping intervention on autistic children's syntactic development and classroom participation within inclusive education settings. The findings indicate that students who received the intervention demonstrated greater improvement in syntactic ability than peers receiving standard instruction, and that gains in syntax were positively associated with higher levels of classroom participation, particularly in communication, engagement and reduced reliance on adult prompting. Importantly, these results clarify a plausible educational mechanism through which language development supports inclusion: syntactic competence appears to function as an enabling resource that facilitates access to instructional discourse and peer interaction. While the findings are necessarily interpreted with caution due to the quasi-experimental design and limited sample size, they nonetheless provide empirically grounded evidence that language-focused interventions can contribute meaningfully to participation outcomes in inclusive classrooms.

The present research offers several important contributions to the literature. First, it advances inclusive education research by empirically linking a specific language intervention to observable classroom participation, moving beyond studies that focus solely on linguistic outcomes. Second, by situating the intervention within authentic classroom contexts and involving teachers as co-implementers, the study strengthens the pedagogical relevance of syntactic bootstrapping as an educational, not merely clinical, approach. Third, the research contributes contextually grounded evidence from an underrepresented educational setting, demonstrating how language interventions can be adapted to local linguistic and cultural conditions. Together, these contributions offer readers a nuanced understanding of how syntactic development intersects with inclusive classroom practice and provide a foundation for future research seeking to refine, extend and contextualize language-based supports for autistic learners in diverse educational environments.

AUTHOR CONTRIBUTIONS

Nur Lailiyah: Conceptualization; investigation; methodology; data curation; writing – original draft. **Anik Lestariningrum:** Project administration; funding acquisition. **Vivi Ratnawati:** Validation; formal analysis. **Meyta Lusiana:** Visualization; supervision. **Farika Nuria Putri:** Software; writing – review and editing.

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CONFLICT OF INTEREST STATEMENT

No conflict of interest found in this research.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available on request from the corresponding author. The data are not publicly available due to privacy or ethical restrictions.

ETHICS STATEMENT

This study received ethical approval from the relevant institutional review board and authorization from the participating schools. Written informed consent was obtained from all parents or legal guardians. Participant anonymity and data confidentiality were strictly maintained throughout the study.

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